

make customers cooperate with utilities to solve their problems. Solutions like C3, AutoGrid, Opower, GE's Grid IQ Insight platform and Siemens' eMeter analytics system can help in this.

2. At the utility level, the AMI data is used for outage management and operational improvements.

3. Solutions from companies like Oracle, SAP, Silver Spring Networks, Siemens/eMeter and ABB/Ventrix help in monitoring transformer health by combining AMI and other grid sensor data with weather and temperature data, asset management system data and other information.

4. With a combination of synchro-phaser deployments, a study of satellite images to predict the next storm, its impact on solar and wind power generation, etc, renewable energy planning systems from grid industry players and IT vendors like IBM help plan distribution better.

5. At the strategic level, the ability to predict is a positive outcome of big data. Of course, the power sector is always

ravaged by the vagaries of nature, and the eccentricities of customers, and it is impossible to accurately know things such as how fast customers will adopt rooftop PV or plug-in electric vehicles, how regulatory and economic developments will impact finances, etc.

Standards bring in order, growth

Karri says, "There are multiple standards governing the space. The key application of these standards we believe and strictly adhere to are in the interoperability space. The kinds of data communication protocol for the energy sector that Symphony Teleca implements include DLMS (preferable) and ANSI C12.22, and when the hardware systems (meters, concentrators, *et al*) are interoperable and software interoperability is needed, we prefer the CIM/ multispeak approach."

In emerging spaces, where there is no clear winner, it can be confusing what standards and protocols to choose for your applications. The

International Electrotechnical Commission's Smart Grid Standards Mapping Tool (<http://smartgridstandardsmap.com/>) can come in handy at such times. It helps you find the best fit for your needs, and covers not just their own standards but all leading ones including IEEE's.

Watch out for smart moves in the power grid

Thakurdesai says, "One of the main trends one can look out for is the integration of renewable energy systems into existing grids catering to higher demand and improving efficiency."

"The world is heading towards a convergence, so we believe there would be an improved involvement of the consumer, taking an increased interest in smart grid cost-benefit analysis, forcing smart grid project leaders to achieve a higher standard for investment justification," adds Karri. ●

The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai

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